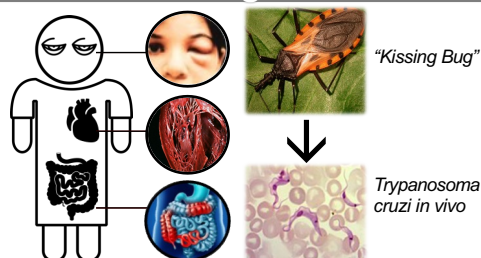


Background



- Chagas Disease (American Trypanosomiasis) is caused by the parasite, *Trypanosoma cruzi*, which is transmitted to humans via the feces of the Triatome bug endemic to Latin America.
- Mortality rate is high, killing around 100,000 individuals annually. The CDC estimates that more than 300,000 people living in the US have Chagas Disease.¹
- Symptoms of the acute stage include fever, vomiting, diarrhea, and inflammatory responses like Romana's sign (a swollen eyelid).
- The chronic stage of Chagas disease is what makes it lethal and includes cardiomyopathy, megacolon, and heart failure.²
- While prevention methods exist, current treatment is inadequate.

Aims

- Design and synthesize novel compounds with high efficacy and specificity at killing the *T. cruzi* parasite
- Compounds should be orally bioavailable, metabolically stable and have a low potential for adverse effects

Methods

- Designed** and **optimized** anti-chagas pharmacophores by developing structure activity relationships using the predictive modeling chemistry software, StarDrop™ (Figure 1)
- Synthesized** compounds via reductive amination pathway
- Purified** and **isolated** target compounds
- The final compound structure verified via ¹H NMR and ¹³C NMR
- Therapeutic **efficacy tested in vivo** (pIC50) by the DNDi

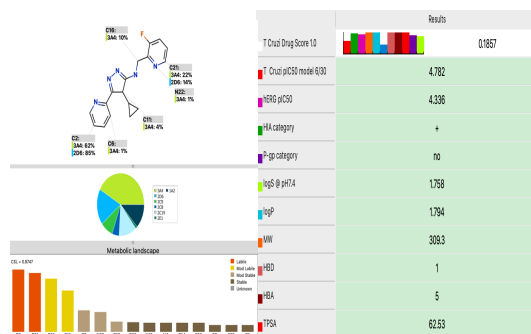
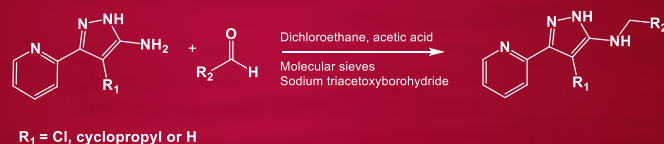


Figure 1: StarDrop predictive models and metabolic profile for compound C



Design, Synthesis, and Assay of Novel Compounds for the Treatment of Chagas Disease

Major pathway used: reductive amination



Peter Shevlin, Brian Chan, Sumin Kim
(PharmD Candidates, 2023)
Chase Smith, PhD and Carolyn Friel, RPh, PhD

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Results



- A StarDrop™ software model was developed to design and predict *T. cruzi* activity of novel compounds (Figure 2).
- Nine compounds were successfully synthesized; yields ranged from 5.3% to 89.7%
- Biological assays showed that all compounds had low to moderate efficacy and excellent safety profiles. Three compounds are pending DNDi assays.

Compound Structure	Efficacy		Toxicity		Yield (%)
	pIC50 predicted	pIC50 actual	MCR5 CC50	PMM CC50	
A	4.8	5.0	35.1	64.0	89.7%
B	4.3	4.6	42.8	64.0	5.3%
C	4.5	4.5	64.0	64.0	7.42%
D	4.5	4.5	64.0	64.0	39.7%
E	4.7	4.5	64.0	64.0	22.3%
F	4.7	4.5	62.4	64.0	53.3%
G	4.7	Pending	Pending	Pending	18.4%
H	4.6	Pending	Pending	Pending	10.1%
I	5.6	Pending	Pending	Pending	11.6%

Figure 2: StarDrop predictive pIC50 values compared to lab measured pIC50 *T. cruzi* values as well as measured MCR5 and PMM CC50 safety values

Discussion & Conclusion

- StarDrop was successful in predicting efficacy and safety profiles of the compounds.
- Reductive amination was an effective synthesis pathway as confirmed by ¹H NMR, ¹³C NMR and mass spectrometry.
- Future laboratory direction is to synthesize additional compounds with cyclic aliphatic substituents that are safe, potent and specific in killing *T. cruzi*

References

- CDC - Chagas Disease - Epidemiology & Risk Factors. Centers for Disease Control and Prevention. <https://www.cdc.gov/parasites/chagas/epi.html>. Published April 11, 2022. Accessed October 27, 2022.
- Chagas disease. Mayo Clinic. <https://www.mayoclinic.org/diseases-conditions/chagas-disease/symptoms-causes/syc-20356212?text=1&from=chagas%20disease%20progress%20to%20heart%20year%20body%20needs>. Published November 12, 2020. Accessed October 27, 2022.

The authors have no financial disclosures



Purpose

Chagas disease is an infection of the parasite, *Trypanosoma cruzi*, and is endemic to Latin America. The chronic stages of Chagas disease are what make it so lethal, resulting in cardiomyopathy. Current first line treatments for Chagas disease are insufficient. Treatment includes antiparasitic agents, such as benznidazole and nifurtimox that are nonspecific for *Trypanosoma cruzi*. These agents are not always effective and have significant toxicity. Therefore, the goal of this project is to design and synthesize orally bioavailable, novel compounds that have high efficacy, specificity, metabolic stability and a low potential for adverse effects.

Methods

The core of our novel compounds, containing a pyridine, pyrazole, and cyclopropane (or Cl), was selected based on previous evidence of *Trypanosoma cruzi* potency from the DNDi data set. The drug discovery software, StarDrop, was then used to design several new compounds. Various substituents were added to the pharmacophore in order to find hit compounds. Predictive models such as *T. cruzi* IC₅₀ and hERG pIC₅₀ were used in StarDrop to assess compounds for efficacy and safety.

Six compounds were synthesized using reductive amination. The crude products were purified using HPLC and recrystallization. Compound identification was confirmed by ¹HNMR, ¹³CNMR and mass spectrometry. The purified, crystalized products were sent for biological testing. Biological assays conducted by the DNDi included *L. infantum* and *T. cruzi* IC₅₀ for efficacy as well as PMM CC₅₀ and MRC5 CC₅₀ for toxicity. Current synthesis is focusing on adding aliphatic rings to the core DNDi compound via the same reductive amination pathway to maximize potency.

Results

A StarDrop software model was developed to predict *T. cruzi* activity of novel compounds. Six compounds were successfully synthesized, and yields ranged from 5.3% to 89.7%. Although biological assays showed that the safety profiles of the compounds were excellent, results showed that all compounds had low to moderate efficacy. The *T. Cruzi* pIC₅₀ values for the six initial compounds are as follows 5.02, 4.46, 4.49, 4.58, 4.48, and 4.54.

Conclusion

Stardrop was successful in predicting efficacy and safety profiles of the compounds. Reductive amination was an effective synthesis pathway as confirmed by ¹HNMR, ¹³CNMR and mass spectrometry. The future direction of this project is to synthesize compounds with cyclic aliphatic substituents that are safe, potent and specific in killing *T. cruzi* with a goal of meeting DNDi standards (*T. Cruzi* IC₅₀ < 0.1) in order to follow up with in vivo testing.

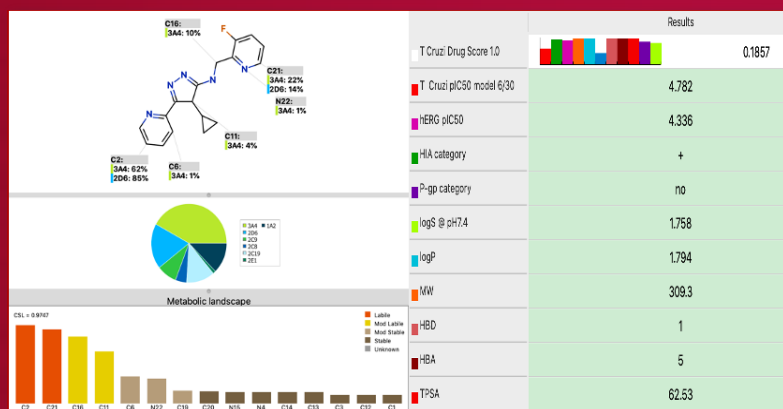


Figure 1: StarDrop predictive models and metabolic profile for compound C

Design, Synthesis, and Assay of Novel Compounds for the Treatment of Chagas Disease



Report integrated screening experiment					
Year	2021	Protocol:	8DIL10 1/4	IC-90 (uM)	
Seq. No.	026 (1)	Date plate production:	2/06/2022		
Exp. No.	2021-026 (1)	Highest dose:	64		
Plate	UA	Supplier-1	Supplier-2	Solubility	IC-90 (uM)
					1 2 3 4 5 6
					MRC-5 Tcruz Linf Tcruc Tcrud PMM
C-340-A3	50118	MCPHS-BC 3-1-A		>	64.00
C-340-B3	50119	MCPHS-BC 4-1-A		>	64.00
C-340-C3	50120	MCPHS-PS 1-1-B		>	64.00
C-340-D3	50121	MCPHS-PS 2-1-C		>	64.00
C-340-E3	50122	MCPHS-SK 2-1-F		>	64.00
C-340-F3	50123	MCPHS-SK 3-1-B		>	64.00

Figure 4: In vitro results obtained in collaboration with the DNDi such as IC50

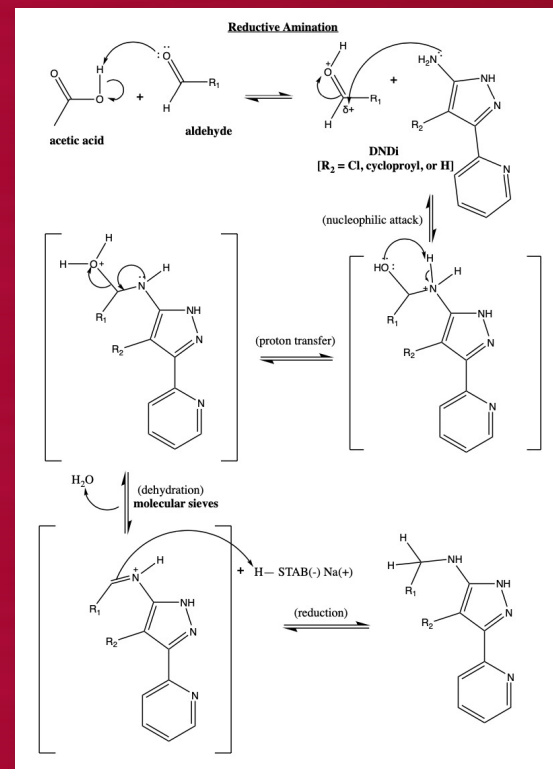


Figure 2: The reaction for the major synthesis pathway performed for all compounds, reductive amination

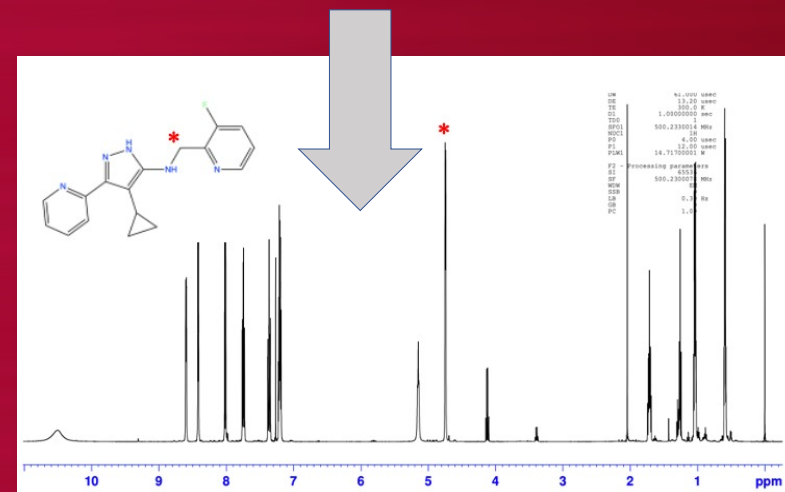


Figure 3: The ¹H NMR obtained for compound C in CDCl₃ showing key identifying peaks such as the singlet at 4.7 ppm. ¹H NMR was the main identifier used.